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Department of Electronics & Communication Engineering | GATE Preparation Library

Comprehensive GATE Notes: Network Theory

Paper Code: EC • Section 2: Networks, Signals and Systems (8-11 Marks)

Subject Code: EC-NET • **Target Exam:** GATE ECE • **Weightage:** 8-11 Marks

1. Network Theorems: Thevenin, Norton & Maximum Power Transfer

- **Thevenin's Theorem:** Any linear bilateral active resistive network can be replaced across terminals A-B by an open-circuit voltage V_th in series with an equivalent resistance R_th.
- **Handling Dependent Sources:** To find R_th with dependent sources: deactivate all independent sources (short voltage sources, open current sources). Connect a test source V_test = 1 V across A-B and find the injected current I_test. Then R_th = V_test / I_test = 1 / I_test.
- **Maximum Power Transfer Theorem:**

• *Pure Resistive:* Maximum power delivered to load R_L when R_L = R_th. Max power P_max = V_th^2 / (4*R_th). Efficiency is exactly 50%.
• *AC Network (variable load Z_L = R_L + jX_L):* Load impedance must be the complex conjugate: Z_L = Z_th^* = R_th - jX_th. Max power P_max = |V_th|^2 / (4*R_th).
• *AC Network (pure resistive load R_L with fixed reactive source):* R_L = |Z_th| = sqrt(R_th^2 + X_th^2).

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Thevenin R_th (Test Method)	**R_{th} = V_{test} / I_{test}**	V_test = 1 V, I_test = Injected Current	Must be used whenever dependent sources exist
Max Power (DC)	**P_{max} = V_{th}^2 / (4 * R_{th})**	R_L = R_th	Occurs when R_L = R_th; 50% power lost internally
Max Power (AC Complex)	**Z_L = Z_{th}^* = R_{th} - j X_{th}**	Z_th = Thevenin Impedance	Reactances cancel (resonance), leaving pure resistive match

GATE PYQ (2 Marks) - AC Maximum Power Transfer
Question: A DC voltage source with open-circuit voltage 24 V has an internal impedance of 6 + j8 Ohms. Find the maximum power delivered to (a) a complex load Z_L, and (b) a pure resistive load R_L.

Step 1: V_th = 24 V, Z_th = 6 + j8 Ohms => R_th = 6, X_th = 8.
Step 2 (Complex Load): Choose Z_L = Z_th^* = 6 - j8 Ohms.
Z_total = 6 + j8 + 6 - j8 = 12 Ohms. Current I = 24 / 12 = 2 A.
P_max = I^2 * R_L = 2^2 * 6 = **24.0 W**. [Formula: 24^2 / (4*6) = 24.0 W].
Step 3 (Resistive Load): R_L = |Z_th| = sqrt(6^2 + 8^2) = 10 Ohms.
Current I = 24 / |(6 + 10) + j8| = 24 / sqrt(16^2 + 8^2) = 24 / sqrt(320) = 1.3416 A.
Power P = I^2 * R_L = (576 / 320) * 10 = **18.0 W**.

2. Transient Analysis: First and Second Order Circuits

- **Continuity Conditions:** Current through an inductor cannot change instantaneously: i_L(0+) = i_L(0-). Voltage across a capacitor cannot change instantaneously: v_C(0+) = v_C(0-).
- **General First-Order Time Response:** For any DC switched first-order RC or RL circuit:
  x(t) = x(inf) + [x(0+) - x(inf)] * e^{-t/tau}

where tau = R_eq * C for RC circuits, and tau = L / R_eq for RL circuits.

- **Second-Order Series RLC Circuit:** Char eq: s^2 + (R/L)s + 1/(LC) = 0 => s^2 + 2*zeta*omega_n*s + omega_n^2 = 0.
- Undamped: zeta = 0 (R = 0)
- Underdamped: zeta < 1 => R < 2*sqrt(L/C)
- Critically Damped: zeta = 1 => R = 2*sqrt(L/C)
- Overdamped: zeta > 1 => R > 2*sqrt(L/C).

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
First Order Step Response	**x(t) = x(inf) + [x(0+) - x(inf)] e^{-t/tau}**	tau = Time Constant	Universal 3-step formula for RC and RL switching
RC Time Constant	**tau = R_{eq} * C**	R_eq = Thevenin Resistance across C	R_eq evaluated looking into open capacitor terminals
RL Time Constant	**tau = L / R_{eq}**	R_eq = Thevenin Resistance across L	R_eq evaluated looking into open inductor terminals

GATE PYQ (2 Marks) - First-Order RC Transient Response
Question: In a series RC circuit with R = 10 kOhms and C = 100 muF, capacitor is initially charged to 5 V. At t = 0, a 20 V DC source is switched in series. Find the voltage across the capacitor at t = 1 second.

Step 1: v_C(0+) = v_C(0-) = 5 V.
Step 2: v_C(inf) = 20 V.
Step 3: tau = R * C = 10,000 * (100 * 10^-6) = 1.0 second.
Step 4: v_C(t) = 20 + [5 - 20] * e^{-t/1.0} = 20 - 15 * e^{-t}.
Step 5: At t = 1 s: v_C(1) = 20 - 15 * e^-1 = 20 - 15 * 0.36788 = **14.48 V**.

3. Two-Port Network Parameters & Reciprocity

- **Z-Parameters (Impedance):** V_1 = z_11 I_1 + z_12 I_2, V_2 = z_21 I_1 + z_22 I_2. Reciprocal iff z_12 = z_21. Symmetric iff z_11 = z_22.
- **Y-Parameters (Admittance):** I_1 = y_11 V_1 + y_12 V_2, I_2 = y_21 V_1 + y_22 V_2. Reciprocal iff y_12 = y_21. Symmetric iff y_11 = y_22.
- **ABCD (Transmission) Parameters:** V_1 = A V_2 - B I_2, I_1 = C V_2 - D I_2. Reciprocal iff AD - BC = 1. Symmetric iff A = D.

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Z Reciprocity / Symmetry	**z_{12} = z_{21} | z_{11} = z_{22}**	z_ij = Impedance Params	Equal cross-terms for reciprocal; equal self-terms for symmetric
ABCD Reciprocity / Symmetry	**AD - BC = 1 | A = D**	A, B, C, D = Cascade Params	Determinant must equal 1 for reciprocal transmission networks
h Reciprocity	**h_{12} = -h_{21}**	h_ij = Hybrid Params	Critical check in BJT small-signal models

GATE PYQ (1 Mark) - ABCD Reciprocity Check
Question: A two-port network has transmission parameters A = 2, B = 15 Ohms, C = 0.1 S. If the network is reciprocal, find parameter D.

Step 1: Reciprocity condition: AD - BC = 1.
Step 2: 2*D - (15 * 0.1) = 1 => 2*D - 1.5 = 1.
Step 3: 2*D = 2.5 => **D = 1.25**.

Essential GATE Exam Pitfalls & Scoring Traps

- **Important:** Never deactivate dependent sources when calculating Thevenin impedance R_th.

- **Important:** For AC maximum power transfer, pay close attention to whether the load is purely resistive or complex.
- **Important:** Inductors act as open-circuits at t = 0+ (if uncharged) and short-circuits at t = inf (DC steady state).